ECMA 30800
Theory of Auctions

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Contents

Preliminaries 1

1.1 Statistics Review

We model information on others' bids as random variables with cumulative distribution functions and densities.

Suppose we have one opponent in the auction. It is important to condition on the event that we win the auction; this is equivalent to conditioning on the event that the other bid is less than yours.

We define the conditional probability density function given $x \leq x_0$ as $\frac{f(x)}{F(x_0)}$ for every $x \leq x_0$.

Now suppose we have k bidders. Order statistics are now important. Suppose we have k values drawn independently from the same density $f(\cdot)$. Let $G_{1,k}$ denote the cumulative density function of the highest among the k values, $G_{2,k}$ the CDF of the second highest, and so on. By definition,

$$\begin{aligned}
G_{1,k}(x) &= \mathbb{P}\left[X_{(1)} \leq x, \ldots, X_{(k)} \leq x\right] \\
&= F^k(x) \\
g_{1,k}(x) &= kF^{k-1}(x)f(x) \\
G_{2,k}(x) &= \mathbb{P}\left[X_{(1)} \leq x, \ldots, X_{(k-1)} \leq x\right] \\
&= \underbrace{F^k(x)}_{X_(k) \leq x} + \underbrace{kF^{k-1}(x)(1-F(x))}_{X_{(k)} > x} \\
g_{2,k}(x) &= k(k-1)F^{k-2}(x)f(x)(1-F(x)).
\end{aligned}$$

1.2 The Four Standard Auctions

In this course, we will mainly consider one-unit auctions in standard contexts. Note that none of these have a reserve price (for now).

Because ties have probability zero (we consider bids on a continuum), we will usually skip consideration of ties.

(i) **First-Price Sealed Bid**: Each bidder submits a sealed bid to the seller. The high bidder wins and pays her bid.

(ii) **Second-Price Sealed Bid**: Each bidder submits a sealed bid to the seller. The high bidder wins and pays the second-highest bid.

(iii) **Dutch**: A public price starts high enough so there are no takers, then decreases over time until someone signals a willingness to pay. They pay the current price.

(iv) **English**: The seller starts with a price of zero and increases it. Each bidder signals when they are no longer willing to pay. Once out, the bidder cannot return. The last remaining bidder wins and pays the current price.

1.3 The Independent Private Values Model

Assuming that private values are independent is very strong; it imposes a restriction on no correlation between values.

Suppose a risk-neutral seller wishes to sell one unit of an indivisible good to one of N risk-neutral bidders. Let v_i denote bidder i's value, known only to her. We assume v_i is drawn independently of others' values from distribution with density f_i.

If player i wins and pays p, her utility is $v_i - p$. If she does not win and pays p, her utility is $-p$.

The Standard Auctions and Their Equilibria 2

To get a solution, we need to impose symmetry on the bidders; i.e., we assume there is one distribution f from which values are drawn.

We standardize to $v_i \in [0, 1]$ and assume each bidder's bid depends on her value. So, a *bidding strategy* is a mapping from values to bids. Because all bidders are ex ante symmetric, we assume that they are using the same bidding function.

2.1 First-Price Auction Equilibrium

Suppose that there is a common distribution of values, so $\forall\ i, f_i = f$.

Definition 2.1.1 (Bid according to a bidding function).

We say that a bidder bids according to a bidding function $b : [0, 1] \to [0, \infty)$ if for any value $v \in [0, 1]$, they bid $b(v)$ when their value is v.

Definition 2.1.2 ((Symmetric) Equilibrium).

A (bidding) function $\hat{b} : [0, 1] \to [0, \infty)$ is a (symmetric) equilibrium of the first-price auction (with symmetric bidders) if each bidder can do no better than to bid according to $\hat{b}$ given everyone else bids according to $\hat{b}$.

Notation 2.1.3.

For any bidder i and $r, v \in [0, 1]$, let $u_i(r, v)$ be bidder i's expected utility when her value is v and her bid is $\hat{b}(r)$ given that all others bid according to $\hat{b}$.

Note that we restrict bids under consideration to those in the range of $\hat{b}$.

Let's assume for now that $\hat{b}$ is strictly increasing in v. Then since player i wins iff $\hat{b}(r) > \hat{b}(v_j)$ for every $j \neq i$ and $\hat{b}$ being strictly increasing implies this happens iff $r > v_j$,

$$\begin{aligned} u_i(r, v) &= \mathbb{P}\left[\text{win} \mid \hat{b}(r)\right](v - \hat{b}(r)) + 0 \\ &= F^{N-1}(r)\left(v - \hat{b}(r)\right) \\ &\leq F^{N-1}(v)\left(v - \hat{b}(v)\right),\ \forall\, r \in [0,1], \end{aligned}$$

where the third equality arises if we assume $\hat{b}$ is an equilibrium.

By this maximization,

$$\begin{aligned} 0 &= (N-1)F^{N-2}(r)f(r)\left(v - \hat{b}(r)\right) - F^{N-1}(r)\hat{b}'(r)\Big|_{r=v} \\ \frac{dF^{N-1}(v)\hat{b}(v)}{dv} &= (N-1)F^{N-2}(v)f(v)v, \quad \forall\, v \in [0,1]. \end{aligned}$$

We can integrate to get

$$F^{N-1}(v)\hat{b}(v) = \int_0^v (N-1)F^{N-2}(x)f(x)x\,dx + C.$$

Plugging in $v = 0$, we see $C = 0$. So,

$$\hat{b}(v) = \frac{N-1}{F^{N-1}(v)}\int_0^v xF^{N-2}(x)f(x)dx.$$

Along this derivation, we assumed that $\hat{b}$ was strictly increasing (Exercise: show it is). We assumed that $\hat{b}$ was an equilibrium; thus, we have only shown uniqueness among strictly increasing bidding functions. It is left as an exercise to show that $\hat{b}$ is indeed an equilibrium. It is left as another exercise to show that uniqueness holds for bidding functions that are not strictly increasing.

Exercise: Suppose f is uniform on $[0,1]$ and find what $\hat{b}$ is.

2.2 Dutch Auction Equilibrium

Dutch auctions are strategically equivalent to first-price auctions; the only relevant consideration to the player in a Dutch auction is at which price she will raise her hand. This is equivalent to a first-price sealed-bid auction because there is no new information gained by nobody having made a bid at a certain point.

The equilibrium is to raise your hand at price $p = \hat{b}(v)$ when your value is v.

2.3 Second-Price Auction

Here, we allow $f_i \neq f_j$ for $i \neq j$.

Bidding your true value is a weakly dominant strategy. To see why, fix

By weakly dominant strategy, we mean that no matter how others bid, you cannot do any better than to bid your true value.

bidder i and let B denote the highest bid of the others. Player i wants to win when $v_i > B$ and wants to lose when $v_i < B$. This is guaranteed by bidding v_i. Choosing a different bid makes the player weakly worse off (strictly when the high bid is near v_i).

Remark 2.3.1. All bids are higher in the second-price auction than the first-price auction — this makes it possible for second-price auctions to raise more revenue.

2.4 English Auction

The English (ascending-price) auction is behaviorally equivalent to the second-price auction.

Remark 2.4.1. In all four auctions we have considered (under the symmetry assumption for the first-price and Dutch auctions), the person with the highest value wins.

2.5 Benchmark Model

Consider $N \geq 2$ bidders with one indivisible object, risk-neutral bidders, private values $v_i \overset{iid}{\sim} F$. We are interested in symmetric equilibria.

Let $Y_1 = \max\{v_j : j \neq i\}$ be the highest opponent value.

In the *first-price* auction, we have the following general characterization.

Proposition 2.5.1 (Equilibrium bidding function, first-price auction).

For general F,

$$b(v) = \mathbb{E}\left[Y_1 \mid Y_1 < v\right].$$

This holds for nonsymmetric value distributions. The bid is equal to the expected highest opponent value, conditional on winning.

2.6 Deriving Closed-Form Solutions

Assume $v_i \sim \text{Unif}[0, 1]$. Then by the above proposition, we see that in the first-price sealed-bid auction (and Dutch auction):

$$\mathbb{P}\left[Y_1 \leq y\right] = y^{N-1}.$$

In particular, with value v, $\mathbb{P}\left[Y_1 < v\right] = v^{N-1}$. Putting g as the density of Y_1,

$$\begin{aligned}
g(y) &= (N-1)y^{N-2} \\
g(y \mid Y_1 < v) &= \frac{g(y)}{\mathbb{P}[Y_1 < v]} = \frac{(N-1)y^{N-2}}{v^{N-1}} \\
b(v) &= \int_0^v y g(y \mid Y_1 < v) dy \\
&= \frac{N-1}{N} v.
\end{aligned}$$

The bidding function is increasing in v. As N grows, players must bid closer to their values.

Revenue Equivalence with Independent Private Values 3

3.1 Direct Comparisons

It's apparent that a first-price and Dutch auction raise the same revenue, and that the second-price and English auctions raise the same revenue.[1] So, we just want to compare the revenue from FPA and SPA.

In a FPA, bidders bid according to

$$\hat{b}(v) = \frac{N-1}{F^{N-1}(v)} \int_0^v x F^{N-2}(x) f(x) dx.$$

Let R_{FPA} be the revenue in a first-price auction and similarly for R_{SPA}.

Let $g_{i,N}(v)$ be the density of the ith highest value v among N values drawn

[1] Here, we really refer to *expected* revenue.

iid from f. Then

$$\begin{aligned}
R_{\text{FPA}} &= \int_0^1 \hat{b}(v) g_{1,N}(v) dv \\
&= \int_0^1 \left[\frac{N-1}{F^{N-1}(v)} \int_0^v x F^{N-2}(x) f(x) dx\right] N f(v) F^{N-1}(v) dv \\
&= N(N-1) \int_0^1 \left[\int_0^v x f(x) F^{N-2}(x) dx\right] f(v) dv \\
&= N(N-1) \int_0^1 \int_0^v f(v) x f(x) F^{N-2}(x) dx dv \\
&= N(N-1) \int_0^1 \int_x^1 f(V) x f(x) F^{N-2}(x) dv dx \text{ because Fubini is dead} \\
&= N(N-1) \int_0^1 x f(x) F^{N-2}(x) \int_x^1 f(v) dv dx \\
&= N(N-1) \int_0^1 x f(x) F^{N-2}(x)(1 - F(x)) dx \\
R_{\text{SPA}} &= \int_0^1 v g_{2,N}(v) dv \\
&= \int_0^1 v N(N-1) f(v)(1 - F(v)) F^{N-2}(v) dv \\
&= N(N-1) \int_0^1 v f(v) F^{N-2}(v)(1 - F(v)) dv.
\end{aligned}$$

First-price and second-price auctions are revenue equivalent, so all four standard auctions raise the same expected revenue (at least when $\forall\ i, F_i = F$).

Proposition 3.1.1.

We can interpret
$$\hat{b}(v_i) = \mathbb{E}\left[\max v_{-i} \mid v = v_{(1)}\right].$$

We condition on $v = v_{(1)}$ since you only win the auction if your bid is highest.

Proof of Proposition 3.1.1. We see

$$\begin{aligned}
\mathbb{E}\left[\max v_{-i} \mid v = v_{(1)}\right] &= \mathbb{E}\left[\max v_{-i} \mid v_{-i} \le v, v = v_{(1)}\right] \\
&= \mathbb{E}\left[\max v_{-i} \mid v_{-i} \le v\right] \text{ by independence.}
\end{aligned}$$

The density of the highest value of others is

$$g_{1,N-1}(x) = (N-1) f(x) F^{N-2}(x).$$

Thus the conditional density given $x \le v$ is $\frac{g_{1,N-1}(x)}{F^{N-1}(v)}$. Now compute

$$\begin{aligned}\mathbb{E}\left[\max v_{-i} \mid v = v_{(1)}\right] &= \frac{1}{F^{N-1}(v)} \int_0^v x(N-1)f(x)F^{N-2}(x)dx \\ &= \frac{N-1}{F^{N-1}(v)} \int_0^v xf(x)F^{N-2}(x)dx \\ &= \hat{b}(v).\end{aligned}$$

□

Corollary 3.1.2.

We have shown that in a first-price auction, a player's bid $b(\hat{v})$ given their value v is the expected second-highest value[a] given that she has the highest value. By symmetry, this is the expected second-highest bid given the highest value is v. This is $R_{\text{SPA}} \mid_{v_{(1)}=v}$.
Clearly, $R_{\text{FPA}} \mid_{v_{(1)}=v} = \hat{b}(v) = R_{\text{SPA}} \mid_{v(1)=v}$. Computing an expectation provides another way of showing revenue equivalence.

[a] Importantly, not bid.

3.2 Direct Selling Mechanisms, Revenue Equivalence

Definition 3.2.1 (Direct selling mechanism).

A DSM is a collection of probability assignment functions $p_i(v_1, \ldots, v_N)$ and cost functions $c_i(v_1, .., v_N)$ such that the following are true.
For each bidder i and vector of values $v_1, \ldots, v_N$, $p_i(v_1, \ldots, v_N) \in [0, 1]$ denotes the probability that bidder i receives the good for sale and $c_i(v_1, \ldots, v_N)$ denotes the cost she pays to the seller.[a] Furthermore, $\sum_i p_i \le 1$.[b]

[a] c_i is paid regardless if player i wins; c_i can be 0.
[b] The seller may keep the good.

A DSM works as following. Bidders privately report values $r_1, \ldots, r_N$, which may be false to the seller, who allocates the good according to $p_1, \ldots, p_N$[2] and the bidders pay their cots $c_1, \ldots .c_N$.

Note that this definition encompasses a very large set of rules for how an "auction" can be run.

While this is a very general framework, we first restrict our consideration to mechanisms where truth-telling is an equilibrium.

[2] A DSM does *not* compute if player i gets the good with probability p_i and also independently allows player j to get the good with probability p_j. Only one player may get the object.

Definition 3.2.2 (Incentive-compatible direct selling mechanisms).

Fix a direct selling mechanism $(p, c)_i$. For each i and each $r_i, v_i \in [0, 1]$, define[a] player i's expected utility when everyone reports their true values v_{-i} and player i reports a value r_i as

$$u_i(r_i, v_i) = \int [p_i(r_i, v_{-i})v_i - c_i(r_i, v_{-i})]\, f_{-i}(v_{-i})\, dv_{-i}.$$

Define[b]

$$\overline{p}_i(r_i) = \int_0^1 \cdots \int_0^1 p_i(r_i, v_{-i}) f_{-i}(v_{-i}) dv_{-i}$$
$$\overline{c}_i(r_i) = \int_0^1 \cdots \int_0^1 c_i(r_i, v_{-i}) f_{-i}(v_{-i}) dv_{-i}$$

as player i's expected probability[c] of winning and cost when she reports a value r_i and the others report their true values v_{-i}. Then

$$u_i(r_i, v_i) = \overline{p}_i(r_i)v_i - \overline{c}_i(r_i).$$

A DSM is incentive-compatible if every bidder i can do no better than to report truthfully given that all other bidders always report truthfully, i.e.

$$v_i = \arg\max_{r_i \in [0,1]} u_i(r_i, v_i) = \arg\max_{r_i \in [0,1]} \overline{p}_i(r_i)v_i - \overline{c}_i(r_i).$$

Truthful reporting is an equilibrium in an incentive-compatible DSM.

[a] Alternatively, derive.
[b] Again, derive.
[c] Expectation from her perspective.

Proposition 3.2.3.

Each of the four standard auctions can be implemented with a direct selling mechanism.

Proof of Proposition 3.2.3. Disregarding the natural ordering, begin with a second-price auction. We need to construct $(p_{\text{SPA}}, c_{\text{SPA}})$. Ignoring ties, define

$$p_i^{\text{SPA}}(v_1, \ldots, v_N) = \begin{cases} 1, & v_i = \max\{v_j\}_j \\ 0, & v < \max\{v_j\}_j \end{cases}$$

and

We define c_i^{SPA} as $v_{(2)}$; by the key property of a SPA, this coincides with $\hat{b}(v_{(2)})$. Importantly, we do *not* use $\hat{b}$ in a definition of c.

$$c_i^{\text{SPA}} = \begin{cases} v_{(2)}, & v_i = \max\{v_j\}_j \\ 0, & v_i < \max\{v_j\}_j \end{cases}.$$

We see that $\left(p^{\text{SPA}}, c^{\text{SPA}}\right)$ is incentive-compatible and implements the outcome of the SPA. Done!

Now consider the English auction. Put

$$\left(p_i^{\text{EA}}, c_i^{\text{EA}}\right) = \left(p_i^{\text{SPA}}, c_i^{\text{SPA}}\right).$$

Mr. Auctioneer can tell the bidders that he will implement their optimal strategies if they tell him their values. Reporting truthfully is a dominant strategy.

For the first-price auction, assume $f_i = f$ for all i. Mr. Auctioneer can tell the bidders the same thing he did as in the English auction with the $\hat{b}$ we derived. Getting scarily close to a proof by tautology, it appears that we can have an auctioneer tell bidders that he will run a certain auction to show that the auction is compatible with a DSM.[3] If all bidders other than i report truthfully, then they all bid according to $\hat{b}$. Thus by our previous derivation player i's dominant strategy is to bid according to $\hat{b}$; i.e. report v_i.

Formally, define

$$p_i^{\text{FPA}}(v_1, \ldots, v_N) = \begin{cases} 1, & v_i = v_{(1)} \\ 0, & v_i < v_{(1)} \end{cases}$$
$$c_i^{\text{FPA}}(v_1, \ldots, v_N) = \begin{cases} \hat{b}(v_{(i)}), & v_i = v_{(1)} \\ 0, & v_i < v_{(1)} \end{cases}.$$

This yields an ICDSM.

Consider the Dutch auction. We are done. □

We have shown the following.

Theorem 3.2.4 (Special Revelation Principle).

For each of the four standard auctions and their equilibria, we can define a DSM (p, c) in which

- Truth-telling in as equiibrium; i.e., (p, c) is incentive-compatible.
- In the truth-telling equilibrium of (p, c), for all i and $(v_1, \ldots, v_N)$, $i's$ winning probability p_i and cost c_i are exactly the same as in the equilibrium of the given standard auction.

Hence, all bidders and the seller are indifferent between the auction and the DSM.

[3] This suffices as a proof for any auction (including the SPA) with an equilibrium [of reporting truthfully] being implementable as a [IC]DSM.

We have not shown the following, which justifies restricting attention to ICDSM when characterizing *equilibrium outcomes* (allocations and payments) that are implementable by some selling mechanism.

Theorem 3.2.5 ((General) Revelation Principle).

For any selling mechanism and any of its equilibria, we can define a DSM in which

- Truth-telling is an equilibrium; i.e., (p, c) is incentive-compatible.
- In the truth-telling equilibrium of (p, c), for all i and $(v_1, \ldots, v_N)$, $i's$ winning probability p_i and cost c_i are exactly the same as in the equilibrium of the selling mechanism.

Hence, all bidders and the seller are indifferent between the selling mechanism and the ICDSM.

Theorem 3.2.6 (Characterizing incentive compatibility).

A DSM (p, c) is incentive-compatible iff for every i,

(i) $\overline{p}_i$ is weakly increasing in v_i on $[0, 1]$ and

(ii) $\overline{c}_i(v_i) = \overline{c}_i(0) + \overline{p}_i(v_i)v_i - \int_0^{v_i} \overline{p}_i(x)dx, \quad \forall\, v_i \in [0, 1]$

Note that (ii) is a strong result; one's expected cost depends entirely on the probability assignment functions, up to a constant (the cost at value 0).

Proof of Theorem 3.2.6. Assume as much differentiability as needed.[4]

($\Longrightarrow$) Recall

$$
\begin{aligned}
u_i(r_i, v_i) &= \overline{p}_i(r_i)v_i - \overline{c}_i(r_i) \\
\text{IC} \implies 0 = \partial_{r_i} u_i(r_i, v_i) \mid_{r_i = v_i} &= \overline{p}_i'(v_i)v_i - \overline{c}_i'(v_i), \quad \forall\, v_i \\
\overline{c}_i(v_i) - \overline{c}_i(v_0) &= \int_0^{v_i} \overline{p}_i'(x)x dx \\
\overline{c}_i(v_i) &= \overline{c}_i(0) + \int_0^{v_i} \overline{p}_i'(x)x dx \\
&= \overline{c}_i(0) + \overline{p}_i(v_i)v_i - \int_0^{v_i} \overline{p}_i(x)dx,
\end{aligned}
$$

where the last equality followed from integrating by parts. We have shown IC $\Longrightarrow$ (ii).

For a maximum at $r_i = v_i$, we require $\frac{d^2}{dr_i^2} u_i(r_i, v_i) \mid_{r_i = v_i} \leq 0$. Hence, we require

$$\overline{p}_i''(r_i)v_i - \overline{c}_i(r_i) \mid_{r_i = v_i} = \overline{p}_i''(v_i)v_i - \overline{c}_i''(v_i) \leq 0$$

[4] Which is actually none.

for every $v_i \in [0,1]$. Differentiating both sides of the FOC from above gives

$$\overline{p}_i''(v_i)v_i + \overline{p}_i'(v_i) = \overline{c}_i''(v_i).$$

Substituting, we see that for a maximum, we require,

$$\overline{p}_i''(v_i)v_i - (\overline{p}_i''(v_i)v_i + \overline{p}_i'(v_i)) \le 0,$$

so $\overline{p}_i(v_i) \ge 0$ for all $v_i \in [0,1]$; we have shown that incentive-compatibility implies $\overline{p}_i$ is nondecreasing.

($\Longleftarrow$) Recall $\frac{d}{dr_i}u_i(r_i,v_i) = \overline{p}_i'(r_i)v_i - \overline{c}_i'(r_i)$. We can differentiate c_i to get

$$c_i'(v_i) = \overline{p}_i(v_i)v_i$$

for every $v_i \in [0,1]$, so

$$\frac{d}{dr_i}u_i(r_i,v_i) = \overline{p}_i'(r_i)(v_i - r_i)$$

which is

$$\begin{cases} \ge 0, & r_i \le v_i, \\ \le 0, & r_i \ge v_i. \end{cases}$$

$\square$

Theorem 3.2.7 (Revenue equivalence).

If two ICDSMs have the same probability assignment functions and every bidder with value zero is indifferent between the two mechanisms, then they each raise the same expected revenue.

Proof of Theorem 3.2.7. Let (p,c) (p,c') be two ICDSMs with the same probability assignment functions. Let R^k be the (expected) revenue from mechanism $k \in \{0,1\}$. Note

Through this proof, we see that indifference at value 0 can be replaced by indifference at any $v \in [0,1]$.

$$\begin{aligned} R^k &= \sum_{i=1}^{n} \int_0^1 \overline{c}_i^k(v_i) f_i(v_i) dv_i \\ &= \sum_{i=1}^{N} \int_0^1 \left[\overline{c}_i^k(0) + \overline{p}_i(v_i)v_i - \int_0^1 \overline{p}_i(x)dx \right] f_i(v_i) dv_i, \end{aligned}$$

where the second equality follows from (ii) from Theorem (3.2.6). Because indifference at value 0 implies

$$\overline{p}_i(0) \cdot 0 - \overline{c}_i^0(0) = \overline{p}_i(0) \cdot) - \overline{c}_i^1(0) \implies \overline{c}_i^0(0) = \overline{c}_i^1(0),$$

we see the right hand side is actually independent of k, so $R^0 = R^1$. $\square$

Remark 3.2.8. We can use the Revenue Equivalence Theorem to solve for $\hat{b}(v)$ in the first-price auction. Suppose $\hat{b}(v)$ is a symmetric equilibrium of a FPA. Then player i's expected cost is $\overline{c}_i(v_i) = F^{N-1}(v)\hat{b}(v)$. By incentive-compatibility, we can plug this into our characterization of ICDSMs and get

$$F^{N-1}(v)\hat{b}(v) = 0 + F^{N-1}(v)v - \int_0^v F^{N-1}(x)dx,$$

so

$$\hat{b}(v) = v - \int_0^v \left(\frac{F(x)}{F(v)}\right)^{N-1} dx.$$

Definition 3.2.9 (Individual rationality).

A DSM (p, c) is individually rational (IR) if for every i and $v_i \in [0, 1]$,

$$\overline{p}_i(v_i)v_i - \overline{c}_i \geq 0.$$

Proposition 3.2.10.

If (p, c) is incentive-compatible, then (p, c) is individually rational iff for all i, $\overline{c}_i(0) \leq 0$.

Proof of Proposition 3.2.10. ($\implies$) Suppose (p, c) is IC and IR. IR gives us $\overline{p}_i(0)\cdot) - \overline{c}_i(0) \geq 0$ for all i, so $\overline{c}_i(0) \leq 0$; incentive computability is not necessary.

($\impliedby$) Suppose (p, c) is IC and $\overline{c}_i(0) \leq 0$ for all i. Then IC gives us the first inequality in

$$\begin{aligned} \overline{p}_i(v_i)v_i - \overline{c}_i(v_i) \geq \overline{p}_i(v_i) & \qquad \geq \overline{p}_i(0)v_i - \overline{c}_i\,(0) \\ \geq -\overline{c}_i(0) \geq 0, & \end{aligned}$$

which holds for all i and v_i. $\square$

A Revenue-Maximizing Auction 4

4.1 Constructing a Revenue-Maximizing Auction

Following Myerson (1981), we can now find the seller's maximum possible revenue among all selling mechanisms that have equilibria by solving the following problem:

Not necessarily a DSM.

$$\max_{(p,c)} \sum_{i=1}^{N} \int_0^1 \overline{c}_i(v_i) f_i(v_i) dv_i$$

such that

(i) $\overline{p}_i(v_i)$ is nondecreasing in $v_i \in [0,1]$ for all i,

(ii) $\overline{c}_i(v_i) = \overline{c}_i(0) + \overline{p}_i(v_i)v_i - \int_0^{v_i} \overline{p}_i(x)dx$, and

(iii) $\overline{c}_i(0) \leq 0$.[1]

Here, (i) and (ii) give us incentive-compatibility, while (iii) gives us individual rationality.

For simplicity, assume that $v_i - \frac{1-F_i(v_i)}{f_i(v_i)}$ is strictly increasing in $v_i \in [0,1]$.

[1]If we have a non-IC DSM where we have an equilibrium, there must be a bidder who wants to report $r_i \neq v_i$. We can turn this into an ICDSM by just changing the probability assignment functions. See the Revelation Principle.

Using (ii) to substitute into the objective, we have

$$\begin{aligned}
R &= \sum_{i=1}^{N} \int_0^1 \left[\overline{c}_i(0) + \overline{p}_i(v_i)v_i - \int_0^{v_i} \overline{p}_i(x)dx\right] f_i(v_i)dv_i \\
&= \sum_{i=1}^{N} \overline{c}_i(0) + \sum_{i=1}^{N} \left[\int_0^1 \overline{p}_i(v_i)v_i f_i(v_i)dv_i - \underbrace{\int_0^1 \int_0^{v_i} \overline{p}_i(x)f_i(v_i)dxdv_i}_{(I)}\right] \\
(I) &= \int_0^1 \int_x^1 \overline{p}_i(x)f_i(v_i)dv_i \\
&= \int_0^1 \left(\int_x^1 f_i(vi)dv_i\right) \overline{p}_i(x)dx \\
&= \int_0^1 (1 - F_i(x))\overline{p_i}(x)dx \\
R &= \sum_{i=1}^{N} \overline{c}_i(0) + \sum_{i=1}^{N} \int_0^1 \phi_i(v_i)\overline{p}_i(v_i)f_i(v_i)\, dv_i \\
&= \sum_{i=1}^{N} \overline{c}_i(0) + \int \left[\sum_{i=1}^{N} \phi_i(v_i)p_i(v)\right] f(v)\, dv.
\end{aligned}$$

where

$$\phi_i(v_i) = v_i - \frac{1 - F_i(v_i)}{f_i(v_i)}.$$

We will find an upper bound for R and then show that this upper bound can actually be achieved. In fact, we will find an upper bound for R ignoring the monotonicity constraint on $\overline{p}_i$, which will produce a bound no lower than the constrained maximum. Note that at this point, the only constraint we still have is (iii),[2] which implies $\sum_i \overline{c}_i(0) \leq 0$.

Looking at the expression for R, we get this upper bound by putting

$$p_i = \mathbb{1}\left(i = \arg\max_j \left(v_j - \frac{1 - F_j(v_j)}{f(v_j)}\right)\right) \mathbb{1}\left(v_i - \frac{1 - F_i(v_i)}{f(v_i)} \geq 0\right).$$

Then

$$R \leq \mathbb{E}\left[\max_i \left[v_i - \frac{1 - F_i(v_i)}{f(v_i)}\right]^+\right].$$

We want to find p^* and c^* such that IC, IR hold, and R achieves this upper bound. We require for each i that

In our function for p_i^*, a tie has probability 0 because the maximand is assumed to be increasing in v_i. Also, the bidder with the highest value need not win with probability 1, since the bidders are not assumed symmetric.

[2] And that $p_i \geq 0$, $\sum_i p_i \leq 1$.

$$\begin{aligned}
\overline{c}_i^*(0) &= 0, \\
p_i^*(v_1, \ldots, v_N) &= \mathbb{1}\left(i = \arg\max_j \left(v_j - \frac{1 - F_j(v_j)}{f(v_j)}\right)\right) \\
&\quad \cdot \mathbb{1}\left(v_i - \frac{1 - F_i(v_i)}{f(v_i)} \geq 0\right).
\end{aligned}$$

With our constraint on $\overline{c}_i^*$, we can get (ii) to be true by defining

$$c_i^*(v_1, \ldots, v_N) = 0 + p_i^*(v_1, \ldots, v_N) - \int_0^{v_i} p_i^*(x, v_{-i}) dx.$$

This (p^*, c^*) achieves the upper bond on revenue is individually rational (satisifes (iii) by construction). It remains to check that this DSM is incentive compatible; i.e., that $\overline{p}_i^*$ is nondecreasing in v_i. Since $v_i - \frac{1-F_i(v_i)}{f(v_i)}$ was assumed to be increasing in v_i and $v_j - \frac{1-F_j(v_j)}{f(v_j)}$ is fixed for $j \neq i$, we see that $p_i^*(v_1, \ldots, v_N)$ is nondecreasing in v_i. Thus, $\overline{p}_i^*$ as an expectation of $p_i^*(v_1, \ldots, v_N)$ is nondecreasing in v_i.

Among all incentive-compatible, individually rational selling mechanisms that have equilibrium, the DSM (p^*, c^*) achieves the upper bound on revenue of

$$R = \mathbb{E}\left[\max_i \left[v_i - \frac{1 - F_i(v_i)}{f(v_i)}\right]^+\right].$$

Remark 4.1.1. We basically showed p^* was unique, while there are multiple ways to choose c^* that yields this maximum revenue, as long as they satisfy (ii) and (iii) in expectation.

Remark 4.1.2. It's possible for a bidder to have a positive value, yet the seller keeps the object for herself. This is because the seller commits ex ante to the mechanism and it's possible for the virtual value to be negative despite a positive value.

Defining the reserve value where the virtual value is zero,

$$r^* : \phi(r^*) = 0 \iff r^* = \frac{1 - F(r^*)}{f(r^*)},$$

we see that if the highest value is below r^*, all virtual values are negative, so the seller ends up assigning winning probability 0 to everyone. In other words, the optimal allocation rule is to sell to the highest bidder if $\max_i v_i \geq r^*$ and keep the good otherwise.

Example 4.1.3 (Second-price auction with reserve price).

Consider an auction where bidders submit bids such that if the highest bid is less than the reserve price r^*, there is no sale, and otherwise, the highest bidder pays $\max\{\text{2nd highest bid}, r^*\}$.

In states where the second-highest bid is low but the highest value is high, the seller earns r^* instead of the second-highest bid, which is lower. If we have two bidders with values $\sim U[0,1]$, then the reserve price is $r^* = 0.5$. There is a tension between the ex ante commitment to the reserve price (which increases expected revenue) and the ex post situation if the second highest value is below the reserve price.
The revenue equivalence theorem does not hold because the probability assignment functions are not the same.

Definition 4.1.4 (Marginal revenue of an auction).

Recall

$$\begin{aligned} R = \int \cdots \int \sum_{i=1}^{N} p_i(v_1, \ldots, v_N) \\ \cdot \left(v_i - \frac{1 - F_i(v_i)}{f(v_i)}\right) f(v) dv \end{aligned}.$$

Define the marginal revenue from bidder i as

$$MR_i(v_i) = v_i - \frac{1 - F_i(v_i)}{f_i(v_i)}.$$

Remark 4.1.5 (Assignment probabilities and marginal revenue). Fix a bidder i with value distribution F_i (density f_i). Consider offering bidder i a take-it-or-leave-it price P. Then the probability of sale is

$$Q(P) \;=\; \Pr[v_i \geq P] \;=\; 1 - F_i(P).$$

Equivalently, for any target sale probability $Q \in [0,1]$, define the *inverse demand* (or quantile-price) function

$$P_i(Q) \;:=\; F_i^{-1}(1 - Q), \qquad \text{so that} \qquad 1 - F_i(P_i(Q)) = Q.$$

If we post price $P_i(Q)$, expected revenue from bidder i is

$$R_i(Q) \;=\; Q \cdot P_i(Q).$$

Define *marginal revenue* as the derivative of this revenue with respect to Q:

$$MR_i(Q) \;:=\; \frac{d}{dQ}\big(QP_i(Q)\big) \;=\; P_i(Q) + Q\,P_i'(Q).$$

To compute $P_i'(Q)$, differentiate the identity $F_i(P_i(Q)) = 1 - Q$:

$$f_i(P_i(Q))\,P_i'(Q) \;=\; -1 \qquad \Rightarrow \qquad P_i'(Q) \;=\; -\frac{1}{f_i(P_i(Q))}.$$

Substituting back gives

$$MR_i(Q) \;=\; P_i(Q) - \frac{Q}{f_i(P_i(Q))}.$$

Now write $v := P_i(Q)$, so that $Q = 1 - F_i(v)$. Then

$$MR_i(Q) \;=\; v - \frac{1 - F_i(v)}{f_i(v)} \;=\; MR_i(v),$$

which matches the virtual-value expression in the definition above.

Remark 4.1.6 (Optimal allocation rule). Because expected revenue can be written as expected surplus of virtual value, the revenue-maximizing allocation is (pointwise in v) to assign the object to the bidder with the highest *nonnegative* marginal revenue:

$$p_i^*(v_1, \ldots, v_N) = \mathbb{1}\left(MR_i(v_i) \geq 0 \;\text{ and }\; MR_i(v_i) > \max_{j \neq i} MR_j(v_j)^+ \right),$$

with the convention that in a tie there is no winner.

In particular, $p_i^*(v) \in \{0, 1\}$ for every profile v, so the mechanism is not all-pay: all bidders with $p_i^*(v) = 0$ receive the object with probability 0, hence (under the standard normalization $c_i^*(0, v_{-i}) = 0$) they pay 0. Therefore, only the winner pays.

Now suppose that $p_i^*(v_1, \ldots, v_N) = 1$. Then

$$\begin{aligned}
c_i^*(v_1, \ldots, v_N) &= p_i^*(v_1, \ldots, v_N)v_i - \int_0^{v_i} p_i^*(x, v_{-i})dx \\
&= 1 \cdot v_i - \int_{v: MR_i(v) = MR_{(2)}}^{v_i} p_i^*(x, v_{-i})dx \\
&= v_i - (v_i - r_i^*) = r_i^*.
\end{aligned}$$

In the second equality, we are saying the lower limit of integration is r_i^*, where

$$\left[r_i^* - \frac{1 - F_i(r_i^*)}{f_i(r_i^*)} \right] = \max_{j \neq i} \left[v_j - \frac{1 - F_j(v_j)}{f_j(v_j)} \right]^+.$$

Existence and uniqueness of r_i^* follows from continuity and strict monotonicity of MR. When bidder i is the winner, she pays r_i^*, which is the highest report[3] bidder i could make without winning (in the case of a tie, there is by definition

[3]This coincides with second-highest value if we know that the auction is incentive-compatible, which is guaranteed for our constructed auction.

of p_i^* no winner).

Since r_i^* does not depend on v_i, the winner's cost does not depend on her reported value. Because $\overline{p}_i$ is increasing in v_i, we see another avenue to show incentive-compatibility.

Remark 4.1.7 (Summary of the constructed optimal auction). Given a vector $(v_1, \ldots, v_N)$ of reports, the good is assigned to the bidder i whose $MR_i(v_i)$ is highest and positive, if any such bidder exists. Losers pay nothing and the winner pays r_i^*, which is the highest bid she could have made without winning.

4.2 Efficiency

Consider an auction with two bidders who have value densities f_1, f_2 and marginal revenue functions $MR_1(v_1), MR_2(v_2)$.

Referring to the graph, a bidder can win even if her value is not the highest; the constructed auction is not necessarily efficient, because value rankings need not be preserved under bidder-specific marginal-revenue maps: in general,

$$v_i > v_j \not\Longrightarrow \quad MR_i(v_i) > MR_j(v_j).$$

If $f = f_1 = f_2$, then $MR_1 = MR_2$, so the auction becomes efficient.

This common increasing MR function induces a unique ρ^* such that

$$0 = \rho^* - \frac{1 - F(\rho^*)}{f(\rho^*)}.$$

Bidders who bid below ρ^* have no chance of winning; among bidders with reported values above ρ^*, the person with the highest reported value wins and pays $\max\{\rho^*, v_2\}$.

Hence, when there is a common value density among all players, the constructed optimal auction is a second-price auction with a reserve price of ρ^*.

Interdependent-Values Correlated Private Information 5

Suppose there is one unit of an indivisible good for sale with N bidders and 1 seller. Bidders do not know their values. Instead, each bidder receives a signal s_i about her value v_i. A bidder's signal is a random variable before it is received, but then deterministic once it is realized.

Let uppercase letters denote random variables and lowercase letter denote their realization. We have the random variables $V_1, \ldots, V_N, S_1, \ldots, S_N$ with joint density $f(v_1, \ldots, v_N, s_1, \ldots, s_N)$.

5.1 Symmetry and Affiliation

Assumption 5.1.1 (Symmetry).

We assume that for every $i \neq j$ and $(v, s) = (v_1, \ldots, v_N, s_1, \ldots, s_N)$,

$$f(v_1, \ldots, v_N, s_1, \ldots, s_N) = f(v'_1, \ldots, v'_N, s'_1, \ldots, s'_N)$$

where $(v', s') = (v'_1, \ldots, v'_N, s'_1, \ldots, s'_N)$ is obtained by substituting v_i, s_i for v_j, s_j from (v, s).
Along this line, we assume that all bidders use the same bidding function.

Assumption 5.1.2 (Signals and values).

We require $s_i, v_i \in [0, \infty)$.

Immediately, this implies

$$\mathbb{E}\left[V_1 \mid S_1 = s_1, S_2 = s_2, \ldots, S_N = s_N\right] = \mathbb{E}\left[V_1 \mid S_1 = s_1, S_2 = s'_2, \ldots, S_N = s'_N\right],$$

where (s') is a permutation of (s).

Note that s_1 does not get swapped with anything, because in the definition of symmetry, if we swap s_i with s_j, we must also swap v_i with v_j.

Remark 5.1.3. The correlated private information auction is a generalization of the independent private values model; we can impose $s_i = v_i$.

Assumption 5.1.4 (Affiliation).

Assume that for all i, $\overline{v} = (\overline{v}_1, \ldots, \overline{v}_N) \geq (\underline{v_1}, \ldots, \underline{v_N}) = \overline{s}$, and $\overline{s} = (\overline{s}_1, \ldots, \overline{s}_N) \geq (\underline{s}_1, \ldots, \underline{s}_N) = \underline{s}$,[a] we have $\frac{f(v_i, \overline{v}_{-i}, \overline{s}_{-i})}{f(v_i, \underline{v}_{-i}, \underline{s}_{-i})}$ is nondecreasing in v_i on $[0, \infty)$ and $\frac{f(\overline{v}, s_i, \overline{s}_{-i})}{f(\underline{v}, s_i, \underline{s}_{-i})}$ is nondecreasing in s_i on $[0, \infty)$ and strictly increasing if $\overline{v}_i > \underline{v}_i$.

[a] Here, the ordering on vectors is defined by the same ordering on all components.

Proposition 5.1.5.

Given these assumptions of symmetry and affiliation,

(i) $\mathbb{E}[V_1 \mid S_1 = s_1, S_2 = s_2, \ldots, S_N = s_N]$ is nondecreasing in $s_2, \ldots, s_N$ and strictly increasing in s_1.

(ii) $\mathbb{E}[\phi(S_1, \ldots, S_N) \mid a_1 \leq S_1 \leq b_1, \ldots, a_N \leq S_N \leq b_N]$ is nondecreasing in $a_1, b_1, \ldots, a_N, b_N$ whenever ϕ is nondecreasing in each of its arguments.

Assumption 5.1.6 (Equilibrium bidding function).

Given symmetry, we also assume that all bidders use the same equilibrium bidding strategy. Furthermore, we assume that bidding is more aggressive when signals are higher. So, one's most serious competitor is the bidder among the others with the highest signal.

This rules out asymmetric equilibria, which certainly exist.

Notation 5.1.7.

We can focus on bidder 1 without loss.
Let $Y_1, Y_2, \ldots, Y_{N-1}$ denote the highest, second-highest, and so on of the bidders' signals $S_2, \ldots, S_N$. Let $X_1, X_2, \ldots, X_{N-1}$ denote the highest, second-highest, and so on of all the signals $S_1, .S_2, \ldots, S_N$.

5.2 Second-Price Auction

Suppose bidder 1 receives a signal s_1. How is she to bid? We conjecture that bidder 1 should bid $\mathbb{E}[V_1 \mid S_1 = s_1]$.

Actually, considering the winner's curse[1], bidding $\mathbb{E}\left[V_1 \mid S_1 = s_1\right]$ is naive and non-strategic.

Definition 5.2.1 (Winner's curse).

Suppose in a second price auction that all bidders bid according to $\tilde{b}(s) = \mathbb{E}\left[V_1 \mid S_1 = s\right]$, which is strictly increasing by affiliation in s. Then the bidder with the highest signal wins. If your signal is s, then $\mathbb{E}\left[V_1 \mid S_1 = s\right] = \mathbb{E}\left[V_1 \mid S_1 = s, 0 \le S_j < \infty \forall\ j \neq 1\right]$. Conditioning on s being the signal that wins,

$$\begin{aligned}\mathbb{E}\left[V_1 \mid S_1 = s = X_1\right] &= \mathbb{E}\left[V_1 \mid S_1 = s,\ 0 \le S_j < s \forall\ j \neq 1\right] \\ &\le \mathbb{E}\left[V_1 \mid S_1 = s_1,\ 0 \le S_j < \infty \forall\ j \neq 1\right] \\ &= \mathbb{E}\left[V_1 \mid S_1 = s\right],\end{aligned}$$

where the inequality is strict if there is actually correlation (usually the case).

There is typically a gap between your bid $\tilde{b}(s)$ and what the good is worth to you conditional on winning.

If the second highest bid is between this gap, the winner of a SPA would pay more than her value if she bid according to $\mathbb{E}\left[V_1 \mid S_1 = s_1\right]$

Let $b^*(s)$ denote the bidding function that all bidders use and suppose $b^*(s)$ is strictly increasing. Assuming that b^* is an equilibrium function, let's find b^* (and later, check that it actually constitutes and equilibrium).

Note that $b^*(y_1)$ is the price bidder 1 pays if she wins. Consider

$$\mathbb{E}\left[V_1 \mid S_1 = s, Y_1 = y_1\right],$$

the expected value conditional on winning. Suppose this has a higher intercept and lower slope than $b^*(y_1)$ due to affiliation.

Then you should bid $b^*(s)$, which is the value of the intersection point of the two functions. We see that $b^*(s) = \mathbb{E}\left[V_1 \mid S_1 = s, Y_1 = s\right]$.

Remark 5.2.2. Strategic bidding is different from estimating the value of the good given your signal:

$$\mathbb{E}\left[V_1 \mid S_1 = s, Y_1 = s\right] \neq \mathbb{E}\left[V_1 \mid S_1 = s\right].$$

[1]If the bid $\mathbb{E}\left[V_i \mid S_i = s_i\right]$ is nondecreasing in s_i, the winner is likely to have gotten a high signal that overestimates the value of the good.

Suppose $s > y_1$, which means that $b^*(s)$ is a wining bid. Then

$$\begin{aligned} b^*(s) = \mathbb{E}[V_1 \mid S_1 = s, Y_1 = s] &\geq \underbrace{\mathbb{E}[V_1 \mid S_1 = s, Y_1 = y_1]}_{\text{correct value estimate}} \\ &> \mathbb{E}[V_1 \mid S_1 = y_1, Y_1 = y_1] \\ &= b^*(y_1). \end{aligned}$$

There is no winner's curse because the correct value estimate is strictly greater than the price paid. (If a bidder wins, she does not regret the bid ex-post.) Now suppose $s < y_1$, so $b^*(s)$ is a losing bid. The

$$\begin{aligned} b^*(s) = \mathbb{E}[V_1 \mid S_1 = s, Y_1 = s] &\leq \underbrace{\mathbb{E}[V_1 \mid S_1 = s, Y_1 = y_1]}_{\text{correct value estimate}} \\ &< \mathbb{E}[V_1 \mid S_1 = y_1, Y_1 = y_1] \\ &= b^*(y_1). \end{aligned}$$

There is no loser's curse because the correct value estimate is less than the price required to win.

Winning bids overestimate the true value, while losing bids underestimate the true value.

Suppose the seller also has private information, namely, a signal S_0. Should the seller reveal or conceal her information? How much should she reveal?

Proposition 5.2.3.

If X and Z have joint density $f(x, z)$, then

$$\mathbb{E}_X[x \mid Z = z] = \int x f(x \mid z)\, dx.$$

This implies

$$\mathbb{E}_Z[\mathbb{E}_X[X \mid Z]] = \mathbb{E}[X].$$

Assume $S_0, S_1, S_2, \ldots, S_n, V_1, \ldots, V_N$ are affiliated and $S_1, S_2, \ldots, S_N$ are ex ante symmetric. For now, consider two possibilities: either the seller commits to always revealing S_0 or to always conceal S_0. If the seller commits to never revealing her information, then the equilibrium is as before: $b^*(s) = \mathbb{E}[V_1 \mid S_1 = s, Y_1 = s]$.

If she commits to always revealing her information, then

$$b^*(s, s_0) = \mathbb{E}[V_1 \mid S_1 = s, Y_1 = s, S_0 = s_0].$$

By the Law of Iterated Expectations,

$$\mathbb{E}[V_1 \mid S_1 = s, Y_1 = s] = \mathbb{E}_{S_0}[\mathbb{E}[V_1 \mid S_1 = s, Y_1 = s, S_0] \mid S_1 = s, Y_1 = s],$$

so

$$b^*(s) = \mathbb{E}_{S_0}\left[b^*(s, S_0) \mid S_1 = s, Y_1 = s\right].$$

Let R_U be the seller's expected revenue when bidders are always uninformed and R_I when they are always informed. Let $X_1 \geq X_2 \geq \cdots \geq X_N$ be the order statistics of $S_1, .S_2, \ldots, S_N$. Then

$$\begin{aligned} R_U &= \mathbb{E}\left[b^*(X_2)\right] \\ R_I &= \mathbb{E}\left[b^*\left(X_2, S_0\right)\right] \\ &= \mathbb{E}_{X_2}\left[\mathbb{E}_{S_0}\left[b^*(S_2, S_0) \mid X_2\right]\right] \\ R_I - R_U &= \mathbb{E}_{X_2}\left[\mathbb{E}_{S_0}\left[b^*(X_2, S_0) \mid X_2\right] - b^*(X_2)\right]. \end{aligned}$$

We conjecture this value is nonnegative. To show it, it suffices to prove that fixing any value $X_2 = s$ guarantees

$$b^*(s) \leq \mathbb{E}_{S_0}\left[b^*\left(s, S_0\right) \mid X_2 = s\right].$$

We see that

$$\begin{aligned} b^*(s) &= \mathbb{E}_{S_0}\left[b^*(s, S_0) \mid S_1 = s, Y_1 = s\right] \\ &= \mathbb{E}_{S_0}\left[b^*(s, S_0) \mid S_1 = X_2 = s, Y_1 = s\right] \text{ since } Y \text{ are the values of the } \textit{others} \\ &\leq \mathbb{E}_{S_0}\left[b^*(s, S_0) \mid S_1 = X_2 = s, Y_1 \geq s\right] \\ &= \mathbb{E}_{S_0}\left[b^*(s, S_0) \mid S_1 = X_2 = s\right] \text{ because } \mathbb{P}\left[Y_1 = s\right] = 0 \\ &= \mathbb{E}_{S_0}\left[b^*\left(s, S_0\right) \mid X_2 = s\right]. \end{aligned}$$

where the last step uses symmetry: conditioning on $S_1 = s$ provides no additional information once $X_2 = s$ is fixed.

For intuition on why the seller should reveal/conceal, first consider the concealment case. If the seller's revenue comes from bidder 1, who bids

$$b^*(s) = \mathbb{E}\left[V_1 \mid S_1 = s, Y_1 = s\right],$$

then this bid being a losing bid (it's the second-highest bid) implies that the $Y_1 > s$ and thus $b^*(s)$ actually underestimates the true value V_1. A more accurate estimate of V_1 would be $\mathbb{E}\left[V_1 \mid S_1 = s, Y_1 \geq s\right] > b^*(s)$. By revealing S_0,, the seller induces bidder 1 to bid $b^*(s, S_0) = \mathbb{E}\left[V_1 \mid S_1 = s, Y_1 = s, S_0\right]$. Because S_0 is affiliated with Y_1 and V_1, conditioning on S_0 will on average help to correct the bias in the estimate of V_1. For losing bids, the bias is downwards, so revealing S_0 tends to increase losing bids (and in particular, the second-highest bid).

Suppose the seller can now *sometimes* reveal her information. We only consider that the case where the bidder reveals if the signal is above α and conceals if it is below. The bidders also know this information, as otherwise, they may think that the seller is always revealing/concealing, and this implies the seller is deceiving the bidders. We won't consider this as a possibility!

We can conjecture that in a FPA, the seller should *not* reveal her information.

By mirroring our previous analysis, when the seller gets a low signal and conceals, the bidders know the signal is low and recondition. But then the seller should reveal her information to raise the second-highest bid, so assuming the seller and bidders have the same understanding of the mechanism, the seller's optimal policy is to always reveal her information.

5.3 English Auction

We will suppose by symmery that all bidders use the same bidding strategy that can be described by the $N-1$ functions $\left(b_0^*, b_1^*, \ldots, b_{N-2}^*\right)$, where given your signal s, $b_0^*(s)$ is the price you drop out if no one has yet dropped out, $b_1^*(s \mid p_1)$ is the price at which you drop out when 1 bidder has dropped out at p_1, and so on, and $b_k^*(s \mid p_1, \ldots, p_k)$ is the price at which you drop out when k bidders have dropped out at $p_1 < p_2 < \cdots < p_k$.

Let's derive these b_k^* functions starting from b_0^*.

Like before, first suppose that $\left(b_0^*, \ldots, b_{N-2}^*\right)$ is a symmetric equilibrium and each $b_k^*(s \mid p_1, \ldots, p_k)$ is strictly increasing in s with the intention to verify it later.

Suppose that all others drop out simultaneously. In this case, they must have the same signal. This is the only scenario that makes b_0^* relevant. Let $y := y_1 = y_2 = \cdots = y_N$. Then $b_0^*(s) = \mathbb{E}\left[V_1 \mid S_1 = s, Y_1 = Y_2 = \cdots = Y_N = y\right]$. Note that $b_0^*(s)$ is strictly increasing.

Suppose one bidder has dropped out first at price p_1. We defined $b_1^*(s \mid p_1)$ as the price at which you drop out when 1 bidder has dropped out at price p_1. Observing p_1 allows all remaining bidders to infer that $Y_{N-1} = y_{N-1}$, where $p_0^*(y_{N-1}) = p_1$.

If all remaining bidders have the same signal y, then

$$b_1^*(s \mid p_1) = \mathbb{E}\left[V_1 \mid S_1 = s, Y_1 = \cdots = Y_{N-2} = s, Y_{N-1} = y_{N-1}\right].$$

While p_1 does not explicitly appear on the right hand side, y_{N-1} is determined by p_1 through $b_0^*(y_{N-1}) = p_1$. We can continue this pattern of proof by graph and find that in general,

$$\begin{aligned} b_k^*(s \mid p_1, \ldots, p_k) = \mathbb{E}\,[V_1 \mid S_1 = s, Y_1 = \cdots = Y_{N-k-1} = s,& \\ Y_{N-k} = y_{N-k}, \ldots, Y_{N-1} = y_{N-1}]&. \end{aligned}$$

Here $b_0^*(y_{N-1}) = p_1$, $b_1^*(y_{N-2}) = p_2$, …, and $b_{k-1}^*(y_{N-k} \mid p_1, \ldots, p_{k-1}) = p_k$.

Does your drop out price ever jump below the clock price when someone else drops out?

Proposition 5.3.1.

A bidder's drop out price never jumps below the clock price when someone else drops out.

Proof of Proposition 5.3.1. We begin by providing intuition for a particular case. Suppose $y_{N-1} < s$.

$$\begin{aligned}
p_1 = b_0^*(y_{N-1}) &= \mathbb{E}\left[V_1 \mid S_1 = Y_1 - \cdots = Y_{n-1} = y_{N-1}\right] \\
&< \mathbb{E}\left[V_1 \mid S_1 = Y_1 = \cdots = Y_{N-2}, Y_{N-1} = y_{N-1}\right] = b_1^*(s_1 \mid p_1) \\
&\le \mathbb{E}\left[V_1 \mid S_1 = Y_1 = \cdots = Y_{N-1} = s\right] \\
&= b_0^*(s).
\end{aligned}$$

Therefore, $p_1 < b_1^*(s \mid p_1) \le b_0^*(s)$.

In general, if $y_{N-1} < y_{N-2} < \cdots < y_{N-k} < s$, then

$$\begin{aligned}
p_k = b_{k-1}^*\left(y_{N-k} \mid p_1, \ldots, p_{k-1}\right) &= \mathbb{E}\Bigg[V_1 \mid S_1 = Y_1 = \cdots = Y_{N-k} = y_{N-k}, \\
&\qquad \underbrace{Y_{N_k+1} = y_{N-k+1}, \ldots, Y_{N-1} = y_{N-1}}_{k-1 \text{ known signals}}\Bigg] \\
&< \mathbb{E}\left[V_1 \mid S_1 = Y_1 \cdots = Y_{N-k-1} = s, \right. \\
&\qquad \left. Y_{N-k} = y_{N-k}, \ldots, Y_{N-1} = y_{N-1}\right] \\
&= b_k^*(s \mid p_1, \ldots, p_k) \\
&\le \mathbb{E}\left[V_1 \mid S_1 = Y_1 = \cdots = Y_{N-k} = s, \right. \\
&\qquad \left. Y_{N-k+1} = y_{N-k+1}, \ldots, Y_{N-1} = y_{N-1}\right] \\
&= b_k^*(s \mid p_1, \ldots, p_k) \\
&= b_{k-1}^*(s \mid p_1, \ldots, p_{k-1}).
\end{aligned}$$

□

Proposition 5.3.2.

$\left(b_0^*, b_1^*, \ldots, b_{N-2}^*\right)$ is a symmetric equilibrium.

Proof of Proposition 5.3.2. We must show that if all others use the b_k^* functions to set their drop-out prices, then you can do no better than do so as well.

Suppose your signal is s. Let the others' signals be $y_1 \ge y_2 \ge \cdots \ge y_{N-1}$. Suppose the second-highest signal is $y_{N-1} < s$. To win, you have to pay

$$b_{N-2}^*\left(y_1 \mid p_1, p_2, \ldots, p_{N-2}\right) = \mathbb{E}\left[V_1 \mid S_1 = Y_1 = y_1, Y_2 = y_2, \ldots, Y_{N-1} = y_{N-1}\right].$$

What is the good worth to you given $y_1 \geq y_2 \geq \cdots \geq y_{N-1}$? It is worth the first term in

$$\begin{aligned}\mathbb{E}\left[V_1 \mid S_1 = s_1, Y_1 = y_1, \ldots, Y_{N-1} = y_{N-1}\right] &\gtreqless \mathbb{E}\left[V_1 \mid S_1 = y_1, \ldots, Y_{N-1} = y_{N-1}\right] \\ &\iff s \gtreqless y_1\end{aligned}.$$

Hence, you want to win exactly when $s > y_1$. You can do this by using the bidding strategy, since drop out prices are increasing in the signal. □

5.4 Revenue Comparison between the Second-Price Auction and English Auction

We conjecture the ascending price will raise more revenue than the second-price auction, since it reveals more information. (Recall that the seller would ex ante always prefer to reveal information she has; so it would make sense for the seller to like others to reveal information as well.)

Let $X_1 \geq X_2 \geq \cdots \geq X_N$ be the order statistics of the N signals $S_1, \ldots, S_N$.

Let SPA$_2$ be a second price actuion in which the seller observes all but the two highest signals and reveals the signals she observes. So, she observes and reveals $X_3 = x_2, \ldots, X_N = x_N$, then allows the bidders with the highest signals to bid in a canoncial SPA. The equilibrium bidding strategy is

$$b^*_{\text{SPA}_2}\left(s \mid x_3, \ldots, x_N\right) = \mathbb{E}\left[V_1 \mid S_1 = Y_1 = s, X_3 = x_3, \ldots, X_n = x_n\right].$$

Therefore, if $X_1 = x_1 \geq X_2 = x_2 \geq \cdots \geq X_N = x_N$, then

$$R_{\text{SPA}_2}(x_1, \ldots, x_N) = \mathbb{E}\left[V_1 \mid S_1 = Y_1 = x_2, Y_2 = x_3, \ldots, Y_{N-1} = x_N\right].$$

In this situation, if we had run an ascending auction instead, then the revenue would be the same. This is because the final drop-out price of the person with the second-highest signal is determined by conditioning on all lower signals. Thus,

$$R_E\left(x_1, \ldots, x_N\right) = R_{\text{SPA}_2}(x_1, \ldots, x_N) \geq R_{\text{SPA}}(x_1, \ldots, x_N),$$

Conjecture 5.4.1. Second-price auctions are used in the real world because of risk aversion.

While the highest signal wins in these auctions, the auctions are not necessarily efficient, because someone with a high value can draw a low signal.

5.5 First-Price Auction

Let $f_{Y_1}(y \mid s)$ be the probability densiity that the highest signal of others is y given that your signal is s. By affiliation,

$$\frac{f_{Y_1}(\overline{y} \mid s)}{f_{Y_1}(y \mid s)}$$

is nondecreasing in s for every $\overline{y} > y$.

This implies that

$$\frac{f_{Y_1}(y \mid s)}{F_{Y_1}(y \mid s)} \tag{5.1}$$

is nondecreasing in s for every y.

Define $w(s, y) = \mathbb{E}[V_1 \mid S_1 = s, Y_1 = y]$ and let $\hat{b}(\cdot)$ be a strictly increasing symmetric equilibrium bidding function (if it exists). Let $u(r, s)$ be your expected utility when you signal is s, you bid $\hat{b}(r)$, and all others bid according to $\hat{b}(\cdot)$. Then, as long as $\hat{b}(\cdot)$ is indeed strictly increasing (which, as before, we plan to check later),

When the highest signal of the others is y, your expected utility is

$$u(r, s) = \int_0^r \left(w(s, y) - \hat{b}(r)\right) f_{Y_1}(y \mid s) dy \tag{5.2}$$

because $\hat{b}$ being strictly increasing implies that $\hat{b}(y) < \hat{b}(r)$ iff $y < r$. Thus, since $\hat{b}$ is an equilibrium, the FOC in r of $u(r, s)$ implies

$$\begin{aligned}
0 = \frac{d}{dr} u(r, s)\bigg|_{r=s} &= \left(w(s, r) - \hat{b}(r)\right) f_{Y_1}(r \mid s) + \int_0^r \left(-\hat{b}'(r)\right) f_{Y_1}(y \mid s) dy \bigg|_{r=s} \\
&= \left(w(s, r) - \hat{b}(r)\right) f_{Y_1}(r \mid s) - \hat{b}'(r) F_{Y_1}(r \mid s) \bigg|_{r=s} \\
\implies \hat{b}'(s) F_{Y_1}(s \mid s) &= \left(w(s, s) - \hat{b}(s)\right) f_{Y_1}(s \mid s).
\end{aligned}$$

The LHS is the marginal cost of increasing your bid, while the RHS is the marginal benefit. We can rearrange to se that for every signal s,

$$\hat{b}'(s) = \left(w(s, s) - \hat{b}(s)\right) \frac{f_{Y_1}(s \mid s)}{F_{Y_1}(s \mid s)} \tag{5.3}$$

We can solve this ODE subject to the boundary condition $\hat{b}(0) = w(0, 0)$:[2]

$$\hat{b}'(s) + \frac{f_Y(s \mid s)}{F_{Y_1}(s \mid s)} \hat{b}(s) = w(s, s) \frac{f_{Y_1}(s \mid s)}{F_{Y_1}(s \mid s)}, \qquad \hat{b}(0) = w(0, 0).$$

[2]We impose this boundary condition because it makes sense over the more sensible $\hat{b}(0) = 0$. For instance, we can give a circular reason. fix.

Define the integrating factor

$$\mu(s) := \exp\left(\int_0^s \frac{f_Y(x \mid x)}{F_{Y_1}(x \mid x)}\,dx\right).$$

Multiplying the ODE by $\mu(s)$ gives

$$\frac{d}{ds}\left(\mu(s)\hat{b}(s)\right) = \mu(s)\,w(s,s)\frac{f_{Y_1}(s \mid s)}{F_{Y_1}(s \mid s)}.$$

Integrating from 0 to s and using $\mu(0) = 1$ yields

$$\mu(s)\hat{b}(s) - \hat{b}(0) = \int_0^s \mu(y)\,w(y,y)\frac{f_{Y_1}(y \mid y)}{F_{Y_1}(y \mid y)}\,dy,$$

so

$$\hat{b}(s) = w(0,0)\exp\left(-\int_0^s \frac{f_Y(x \mid x)}{F_{Y_1}(x \mid x)}\,dx\right) + \int_0^s w(y,y)\,h(y \mid s)dy.$$

Evaluating the integral $\int_\epsilon^s \frac{f_Y(x|x)}{F_{Y_1}(x|x)}dx$, we see that as $\epsilon \downarrow 0$, the integral diverges to ∞, so the first term in $\hat{b}$ is 0. Therefore,

$$\hat{b}(s) = \int_0^s w(y,y)h(y \mid s)dy \tag{5.4}$$

where

$$h(y \mid s) := \frac{f_{Y_1}(y \mid y)}{F_{Y_1}(y \mid y)}\exp\left(-\int_y^s \frac{f_Y(x \mid x)}{F_{Y_1}(x \mid x)}\,dx\right), \qquad 0 \le y \le s.$$

Letting $H(y \mid s) = \int_0^y h(z \mid s)dz$, we get

$$H(y \mid s) = e^{-\int_y^s \frac{f_{Y_1}(x|x)}{F_{Y_1}(x|x)}dx}$$

for every $y \le s$. Thus,

$$H(s \mid s) = 1, \quad H(0 \mid s) = 0,$$

and thus $h(y \mid s) > 0$ is a density on $y \in [0,s]$.

Equation (5.4) is defined only for $s > 0$. This equation says that $\hat{b}(s)$ is a weighted average of $w(y,y)$s with $y \in [0,s]$. Since $w(y,y)$ is strictly increasing in y,

$$\begin{aligned}\hat{b}(s) &= \int_0^s w(y,y)h(y \mid s)dy \\ &< \int_0^s w(s,s)h(y \mid s)dy \\ &= w(s,s)\int_0^s h(y \mid s)dy \\ &= w(s,s)\end{aligned} \tag{5.5}$$

for every s. Combining equations (5.3) and (5.5) yield that $\hat{b}'(s) > 0$ for every s. Therefore, (5.2) is indeed the correct formula for $u(r, s)$ given $\hat{b}(\cdot)$ defined by (5.4).

We have shown that $\hat{b}$ is increasing and that it satsfies the FOC of a maximization problem (by construction). It remains to check that it satisfies the second-order condition:

$$\begin{aligned}\frac{d}{dr}u(r,s) &= \left(w(s,r) - \hat{b}(r)\right) f_{Y_1}(r \mid s) - \hat{b}'(R)F_{Y_1}(r \mid s) \\ &= F_{Y_1}(r \mid s)\left[w(s,r) - \hat{b}(r)\frac{f_{Y_1}(r \mid s)}{f_{Y_1}(r \mid s)} - \hat{b}'(r)\right] \gtreqless 0.\end{aligned}$$

Trying to analyze this in terms of fixing r at s and hypothesizing changes in r, fix $r = s$ and try changing s. By (5.3) and $w(s, r)$ being strictly increasing in s (both from affiliation), we see that decreasing s to below r, the expression is negative, and increasing s to above r, the expression is positive. Therefore, the derivative is maximized in r at $r^* = s$.

Aside 5.5.1. What is the interpretation of $h(y \mid s)$? Reny and Brooks do not know, but Chen may.

Proposition 5.5.2.

We claim that $\frac{f_{Y_1}(y|s)}{F_{Y_1}(s|s)}$ is the density of second highest signal X_2 given that the highest signal X_1 is s.

Proof of Proposition 5.5.2. We know that $f_{Y_1}(y \mid s)$ is the density of $Y_1 = y$ given $S_1 = s$. Thus, $\frac{f_{Y_1}(y|s)}{F_{Y_1}(a|s)}$ is the density of $Y_1 = y$ given $S_1 = s$ and $Y_1 \leq a$. We plug in $a = s$ to conclude, as

$$Y_1 = y_1 \mid S_1 = s, Y_1 \leq s \iff X_2 = y \mid S_1 = s = X_1 \iff X_2 = y \mid X_1 = s.$$

□

Proposition 5.5.3.

The revenues of different auctions are ordered as

$$R_{\text{FPA}} \leq R_{\text{SPA}} \leq R_E.$$

Proof of Proposition 5.5.3. We have already shown the second inequality.

In a second price auction,

$$b^*(s) = \mathbb{E}\left[V_1 \mid S_1 = s, Y_1 = s\right] = w(s,s).$$

Let $R|_{X_1=s}$ be the expected revenue given that the highest signal is s. We see that

$$R_{\text{FPA}}|_{X_1=s} = \hat{b}(s),$$

while

$$R_{\text{SPA}}|_{X_1=s} = \int_0^s w(y,y)\mathbb{P}\left[X_2 = y \mid X_1 = s\right] dy = \int_0^s w(y,y)\frac{f_{y_1}(y \mid s)}{F_{Y_1}(s \mid s)}dy.$$

If we can show that $\hat{b}(s)$ is less than this integral for every s, then we will have shown the revenue of an SPA is greater than that of a FPA.

Recall that we found last time

$$\hat{b}(s) = \int_0^s w(y,y)h(y \mid s)dy.$$

We want to show that $h(y \mid s)$ is greater than $\frac{f_{Y_1}(y|s)}{F_{Y_1}(s|s)}$ for large y. Hence, it suffices to show

$$e^{-\int_y^s \frac{f_{Y_1}(x|x)}{F_{Y_1}(x|x)}dx} = H(y \mid s) \geq \frac{F_{Y_1}(y \mid s)}{F_{Y_1}(s \mid s)}$$

for $y \leq s$. We have that

$$\begin{aligned}
H(y \mid s) &= \exp\left(-\int_y^s \frac{f_{Y_1}(x \mid x)}{F_{Y_1}(x \mid x)}dx\right) \\
&\geq \exp\left(-\int_y^s \frac{F_{Y_1}(x \mid x)}{f_{Y_1}(x \mid x)}dx\right) \text{ since } \frac{f_{Y_1}(a \mid b)}{F_{Y_1}(a \mid b)} \text{ is nondecr. in } b \\
&= \exp\left(-\left(\ln\left(F_{Y_1}(x \mid s)\right)\right)\Big|_y^s\right) \\
&= \exp\left(-\left(\ln\left(\frac{F_{Y_1}(s \mid s)}{F_{Y_1}(y \mid s)}\right)\right)\right) \\
&= \exp\left(\ln\left(\frac{F_{Y_1}(y \mid s)}{F_{Y_1}(s \mid s)}\right)\right) \\
&= \frac{F_{Y_1}(y \mid s)}{F_{Y_1}(s \mid s)}.
\end{aligned}$$

□

Markets and Auctions 6

6.1 Independent Private Values

Consider now m homogeneous units of an indivisible good available (an inelastic supply). There are N bidders, each with single-unit demand. Let $v_i \in [0, 1]$ be i's value for one unit of the good, where v_i has density f_i.

If i's value (for one unit) is v_i, her demand function is

$$Q(p) = \begin{cases} 0, p > v_i \\ 1, p \leq v_i \end{cases} .$$

Let $X_1 \geq \cdots X_N$ be the order statistics of $V_1, \ldots, V_N$. If

$$X_1 = x_1 > X_2 = x_2 > \cdots > X_N = x_n,$$

then the market demand function is

$$Q(p) = \sum_{i=1}^{n} \mathbb{1}_{p \leq X_i}.$$

The market-clearing price is any price $p \in (x_{m+1}, x_{(m)}]$. Note that bidders m and $m + 1$ may have some incentive to not report their values truthfully.

Let b_i denote the "height" of bidder i's submitted one-step demand function; i.e., the price at which she switches from wanting to buy the object vs. not. We will call b_i a bid.

Let $B_{(1)} \geq B_{(2)} \geq \cdots \geq B_{(N)}$ be the order statistics of bids $b_1, b_2, \ldots, b_N$. The range of market-clearing prices is $p \in (B_{(m+1)}, B_{(m)}]$.

We wan to set a rule for the market-clearing price. We shall always choose the highest market-clearing price: namely, $B_{(m+1)}$, which is the $(m+1)$st-highest bid. With this rule, the bidders submit a (sealed) bid.

If $m = 1$, this is a second-price auction.

Proposition 6.1.1.

It is a dominant strategy to bid your value.

Proof of Proposition 6.1.1. We notate $B^{-i}_{(k)}$ as the kth highest bid of the bidders other than i. If bidder i wins a good, she pays $B^{-i}_{(m)}$. You want to win when $v_i > B^{-i}_{(m)}$ and lose when $v_i < B^{-i}_{(m)}$. Since your bid does not affect $B^{-i}_{(m)}$, then bidder i cannot affect the price, then we can accomplish this by bidding $b_i = v_i$. $\square$

If instead $p = B_{(m)}$, a bidder can affect the clearing price directly, so truthful bidding is no longer dominant.

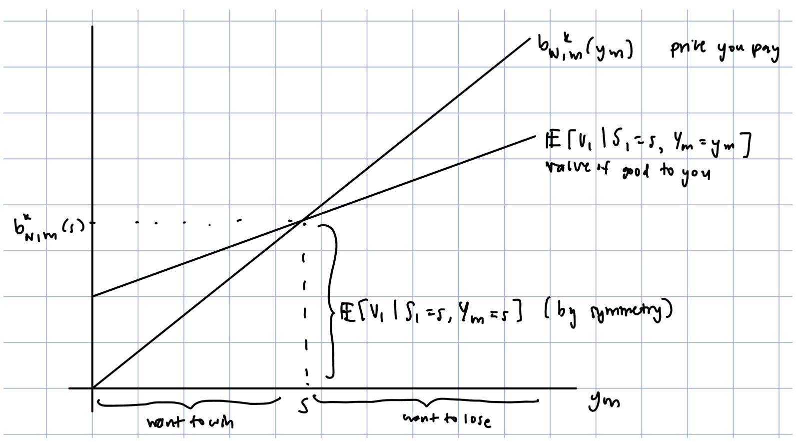
Remark 6.1.2. In this equilibrium, the outcome is efficient.

6.2 Interdependent Values, Affiliated Private Information

Return to the general symmetric interdependent-values model with values $V_1, \dots, V_n$ and signals $S_1, \dots, S_n$. Now consider $m \geq 1$ homogeneous units of a good for sale with at most single-unit demand. Consider an $(m+1)$st price auction. Let $b^*_{N,m}(s)$ be a strictly increasing symmetric equilibrium bidding function.

Fix bidder i (WLOG consider $i = 1$) and let $y_1 > y_2 > \cdots > y_{N-1}$ represent the signals of the other bidders. Then if bidder i wins the good, she pays $b^*_{N,m}(y_m)$. When her signal is s, the good is worth $\mathbb{E}\left[V_1 \mid S_1 - s, Y_m = y_m\right]$.

Figure 6.1: Optimal bidding function against expected value of good.



Considering the picture, the optimal bidding function is

$$b^*_{N,m}(s) = \mathbb{E}\left[V_1 \mid S_1 = s, Y_m = s\right].$$

When does a market price effectively aggregate and reflect private information? If the market is sufficiently large (with the intention of using the LLN), will the market price reflect the true value[1] of the good? We will specialize our model so that $V_1 = \cdots = V_n = V$ and that for any given value $v \in [0, 1]$, each bidder receives an independent signal drawn from $\mathrm{Uni}(0, v)$.[2] Lastly, suppose that v is drawn according to the density $g(v)$ with support $[0, 1]$.

Suppose $N, m \to \infty$ such that $\frac{m}{N} \to \alpha \in (0, 1)$. Since $S \sim \mathrm{Uni}(0, v)$, writing $X_1 \geq X_2 \geq \cdots \geq X_N$ as the order statistics of the signals $S_1, \ldots, S_N$, we expect

$$X_{m+1} \to (1 - \alpha)v.$$

For any bidder whose signal is s,

$$b^*_{N,m}(s) = \mathbb{E}\left[V_1 \mid S_1 = s, Y_m = s\right] \approx \mathbb{E}\left[V_1 \mid X_{m+1} = s\right],$$

so as $N, m \to \infty$ where $\frac{m}{N} \to \alpha \in (0, 1)$, we have that

$$\frac{X_{m+1}}{1 - \alpha} \to v.$$

Hence, for large N, m,

$$b^*_{N,m}(s) \approx \frac{s}{1 - \alpha}.$$

This implies that the price when there are N bidders and m units of the good is

$$p_{N,m} = b^*_{n,m}(X_{m+1}) \approx \frac{X_{m+1}}{1 - \alpha} \to \frac{(1 - \alpha)v}{1 - \alpha} = v.$$

6.3 Multi-Unit Demand

Suppose we have m homogeneous units of the good for sale, N bidders, and private values for each bidder i of $v_i = (v_{i,1}, \ldots, v_{i,m})$ such that $v_{i,1} \geq v_{i,2} \geq \cdots \geq v_{i,m}$; in other words, we have demand decreasing in the number of goods bidder i ends up with. We take $v_1, \ldots, v_N \in [0, 1]^m$ drawn independently from $\{f_i(v_i)\}$. Each bidder i will submit m bids $b_i = (b_{i,1}, \ldots, b_{i,m})$.

What are the analogues of the FPA, SPA, and English auctions? Are there auctions that allocate the m units efficiently?

By efficiently, we mean that after an efficient auction, no pair of bidders would want to buy/sell some unit of the good after the conclusion of the auction. In other words, all $v_{i,k}$ associated with the kth unit allocated to bidder i must be higher than all $v_{j,l}$ for when bidder j did not receive a lth unit.

Note that $v_{i,k}$ is bidder i's *marginal* value for a kth unit of the good. So, if bidder i wins k units and pays a total price of p, her utility is $v_{i,1} + \cdots + v_{i,k} - p$.

Consider the $(m + 1)$*st price auction*[3] as a generalization of the second price auction.

[1] Presumed to be a common value as in stocks, treasury bonds, IPOs, etc.

[2] The distribution need not be uniform. We just require that $\frac{f(\bar{s}|v)}{f(s|v)}$ is increasing in v for $\bar{s} > s$. Equivalently, we can require that for $v_2 > v_1$, $\frac{f(s|v_2)}{f(s|v_1)}$ is increasing in s.

[3] Also called the uniform-price auction.

Definition 6.3.1 ((m+1)st-price auction (Uniform-price auction)).

Each bidder i submits $b_i = (b_{i,1}, \dots, b_{i,m})$. Without loss we can take $b_{i,1} \geq \dots \geq b_{i,m}$. The seller orders all $N \cdot m$ bids and the m highest bids win. If bidder i wins k units of the good, then she pays the $(m+1)$st-highest bid for each unit won, so she pays $kB_{(m+1)}$.

Is the outcome always efficient in an $(m+1)$st-price auction?

Example 6.3.2.

Let $N = m = 2$ with $v_1 = (10, 10)$ and $v_2 = (7, 0)$. Each bidder submits two bids, and the two units of the good go to bidder 1 in an efficient allocation. Under honest bidding, this is an equilibrium. But under strategic bidding, bidder 1 could bid $(10, 0)$ and be better off.
The uniform-price auction does not lead to an efficient outcome because one can affect the $B_{(m+1)}$ bid, unlike as in the single-unit second price auction. A bidder with high values for multiple units may strategically bid low on later units to still win (some) units but pay a lower price for them.

What if we consider a *sequential second-price auction*, in which one good is auctioned off at a time? In the previous example, honest bidding is an equilibrium, but in strategic bidding, bidder 1 could first bid 0, then 10, to increase her utility.

Recall that in the single-unit SPA, if a bidder is the one who sets the price, then she does not win the good.

Consider elettering the goods $A, B, \dots$ and run English auctions for the goods with separate price clocks. Each price clock will increase until demand for the good is 1. With the same example as before, honest bidding is efficient, while strategic bidding is not. This is because bidder 1 could commit to only bidding on unit A. Then bidder 2 gets unit B for price 0, and bidder 1 gets unit A for price 0.

The prices are 0 and not some small ϵ (or 1 if we have discrete prices) as long as the bidders start the bidding on different goods.

Let us consider the *discriminatory auction*. Suppose each bidder submits $b_i - (b_{i,1}, \dots, b_{i,m})$ and the highest m bids win, where winners pay their bids on each unit won. If i wins k units, then

$$u_i - v_{i,1} + \dots + v_{i,k} - (b_{i,1} + \dots + b_{i,k})\,.$$

Suppose $N = m = 2$ and let $(\hat{b}_1, \hat{b}_2)$ denote the equilibrium bidding strategy for the discriminatory auction in which $f_i = f$ for all i. Here, $\hat{b}_k(v_{i,1}, v_{i,2})$ is the bid on a kth unit submitted by i when $v_i = (v_{i,1}, v_{i,2})$.

Assuming that $\hat{b}_k$ is strictly increasing in $v_{i,1}$ and $v_{i,2}$.

What would this bidding function look like? Laboriously, we find that when $v_{i,1} \approx v_{i,2}$, because i's bid for a first unit is against the lower bid distribution of $\hat{b}_2\,(v_{j,1}, v_{j,2})$ than i's bid for a second unit against the higher distribution

$\hat{b}_1(v_{j,1}, v_{j,2})$ is, we reason that *ideally* i would want $b_{i,1} \le b_{i,2}$.

This auction is not efficient with positive probability.

Maybe it would be a good idea to characterize exactly when it is not efficient.

6.4 Vickrey's Auction

We want an auction that allocates the m units to the bidders with the highest m vlaues. If we can get the bidders to be honestly such that

$$b_i = (v_{i,1}, \dots, v_{i,m}), \quad \forall\ i,$$

then allocating the units to the m highest bids will be efficient. So, let's focus on sealed-bid auctions that allocated units to the m highest bids. The remaining task is to determine a payment structure that is incentive-compatible. We will assume for (and verify later that such a design is possible) that one will only pay if she wins some units.

Is there an auction design that is not incentive-compatible but still has bids increasing in values, so it is efficient? (Besides some case where you pay α times the bid we derive, so you bid $\frac{v}{\alpha}$.)

Suppose $N = m = 2$, and suppose bidders 2 bids $b_2 = (b_{2,1}, b_{2,2})$, with $b_{2,1} > b_{2,2}$. Can we define a payment rule for bidder 1 that makes him win when he wants ot win and lose when he wants to lose when he bids honestly?

Let $p_{1,1}$ and $p_{1,2}$ be the prices bidder 1 would pay for the first and second unit. If bidder bids honestly, he wins at least one unit iff $v_{1,1} > b_{1,1} > b_{2,2}$ and wins a second unit iff $v_{1,2} = b_{1,2} > b_{2,1}$. He wants to win a first unit when $v_{1,1} > p_{1,1}$ and a second unit when $v_{1,2} > p_{1,2}$. Then we can set $p_{1,1} = b_{2,2}$ so the first-unit condition is satisfied, and set $p_{1,2} = b_{2,1}$ so the second-unit condition is satisfied.

Note that bidder i cannot influence the prices she pays, and over/underbidding can only result in winning a unit when she doesn't want to/wants to win.

We write this generically as the pricing rule

$$p_{i1} = b_{j2}, \quad p_{i2} = b_{j1}, i \neq j.$$

Under this pricing rule, bidder i only pays for units she receives, as we assumed.

Now consider $N = 2$ and m units. If bidder 1 bids honestly, so

$$b_1 = v_1 = (v_{1,1}, \dots, v_{1,m}),$$

and bidder 2's bids are fixed at $b_2 = (b_{2,1}, \dots, b_{2,m})$, then

- Bidder 1 wins at least one good if $b_{1,1} = v_{1,1} < b_{2,m}$,
- Bidder 1 wins at least two goods if $b_{1,2} = v_{1,2} < b_{2,m-1}$,
- and so on.

Hence, $v_{1,1}$ competes against $b_{2,m}$, $v_{1,2}$ competes aaginst $b_{2,m-1}$, $v_{1,3}$ competes against $b_{2,m-2}$, and so long.

Analogously,

- Bidder 1 wants to win at least one good if $v_{1,1} > p_{1,1}$,
- Bidder 1 wants to win at least two goods iff $v_{1,2} > p_{1,2}$,

- and so on.

Therefore, we should put

$$p_{1,k} = b_{2m-k+1}.$$

Remark 6.4.1. We can check incentive-compatibility by asking whether honest bidding admits any unilateral profitable deviation. Suppose $b_i = (v_{i_1}, \ldots, v_{i,m})$ and bidder i wants to win k units, while bidder j bids $b_j = (b_{j,1}, \ldots, b_{j,m})$. Then $v_{i,1}, v_{i,2}, \ldots, v_{i,k}$ and $b_{j,1}, b_{j,2}, \ldots, b_{j,m-k}$ are the m highest bids.

Would bidder i have liked to submit a different bid vector and win a different number of units? For a $k+1$st good, bidding i must pay $b_{j,m-k}$, which by being in the highest m bids, is larger than $v_{i,k+1}$, which is not in the highest m bids. The same applies to the values vs. prices for the $k+2$nd good and so one. Similarly, the kth unit that bidder i won has value $v_{i,k} > b_{j,m-k} = p_{i,k}$, and so on for the $k-1$st and earlier units.

We note that bidder i, having won k units, pays the sum of the k losing bids of his opponent. The same pricing rule applies to the second-price auction, so this generalizes the SPA.

Now suppose we have N bidders and m units.

Definition 6.4.2 (Vickrey auction).

Suppose N bidders submit m sealed bids and the m highest bids win. If a bidder wins k units, then she pays the sum of the k highest losing bids of her opponents.

Theorem 6.4.3 (Efficiency of Vickrey Auction).

Bidding honestly is a dominant strategy in a Vickey auction. Thus, the outcome of rational bidding is efficient.

Proof of Theorem 6.4.3. If bidder i bids truthfully with $b_i = (v_{i,1}, \ldots, v_{i,m})$ and bidders $j \neq i$ bid $(b_{j,1}, \ldots, b_{j,m})$. Let $B^{-i}_{(k)}$ be the kth highest bid of the others. Put the price i pays for a kth unit be $p_{i,k}$. Bidder i wins a first unit iff $v_{i,1} > B^{-i}_{(m)}$, a second unit iff $v_{i,2} > B^{-i}_{(m-1)}$, and so on.

The reasoning for the proof is similar to the $N = 2$ case we already saw. $\square$

6.5 Efficiency and Interdependent Values, Multiple Units and Multi-Unit Demand in the Vickrey Auction

We specialize to $N = 2$ and m units. Each bidder $i = 1, 2$ receives a signal s_i. Let $v_{i,k}(s_1, s_2)$ denote i's value for a kth unit when the signals are $s_1, s_2 \in [0, \infty)$.

For example, we may have

$$v_{i,k}(s_1, s_2) = \mathbb{E}\left[V_{i,k} \mid S_1 = s_1, S_2 = s_2\right],$$

but we will make assumptions on the values of marginal goods $v_{i,k}(s_1, s_2)$ directly.

Assumption 6.5.1.

We assume for every s_1, s_2 and $k, l, i \neq j$,

(i) $v_{i,k}(s_1, s_2) \geq v_{i,k+1}(s_1, s_2) \geq 0$

(ii) $v_{i,k}(s_1, s_2)$ is strictly increasing in s_i and nondecreasing in s_j

(iii) The partials satisfy

$$\frac{\partial v_{1,k}(s_1, s_2)}{\partial s_1} \geq \frac{\partial v_{2,l}(s_1, s_2)}{\partial s_1}, \quad \frac{\partial v_{1,k}(S_1, s_2)}{\partial s_2} \leq \frac{\partial v_{2,l}(s_1, s_2)}{\partial s_2}.$$

(iv) There exist unique α, β such that

$$v_{i,k}(s_1, \alpha) = v_{2,l}(s_1, \alpha), \quad v_{1,k}(\beta, s_2) = v_{2,l}(\beta, s_2).$$

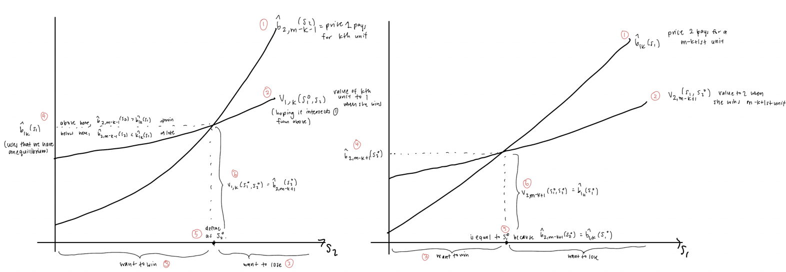
This assumption is stronger than what's needed but simplifies things.

Let $\hat{b}_i(s_i) = \left(\hat{b}_{i,i}(s_i), \ldots, \hat{b}_{i,m}(s_i)\right)$ for $i \in \{1, 2\}$ be the equilibrium bidding vectors when the bidders have signals s_1 and s_2. Assume all $\hat{b}_{i,k}(s_i)$ are increasing in s_i.

As we know $\hat{b}_{1,k}(s_1)$ competes against $\hat{b}_{2,m-k+1}(s_2)$ and vice versa. Also, $\hat{b}_{2,m-k+1}(s_2)$ is the marginal price that bidder 1 pays for the kth unit she wins and $\hat{b}_{1,k}(s_1)$ is the marginal price that bidder 2 pays for his $m-k+1$st unit. Hence, bidder 1's optimal bid for a kth unit depends on $\hat{b}_{2,m-k+1}(s_2)$ and bidder 2's optimal bid for a $m-k+1$st unit depends on $\hat{b}_{1,k}(s_1)$. We need to derive these together, as each function appears in the other's first-order condition.

Let s_1^0 be any signal for bidder 1. We'd like to determine $\hat{b}_{1,k}(s_1^0)$.

Figure 6.2: Derivation of the paired bidding functions.



The graphs each depict an equilibrium condition: at the marginal threshold

(s_1^0, s_2^0) where bidder 1's bid for unit k exactly ties bidder 2's competing bid for unit $m-k+1$, each bidder optimally bids their true value at that threshold. Note that from the first graph, we have the equation

$$\hat{b}_{1,k}(s_1^0) = v_{1,k}(s_1^0, s_2^0) = \hat{b}_{2,m-k+1}(s_2^0).$$

Furthermore, from the second graph,

$$\hat{b}_{2,m-k+1}(s_2^0) = v_{2,m-k+1}(s_1^0, s_2^0) = \hat{b}_{1,k}(s_1^0).$$

Since both conditions hold simultaneously, combining them gives

$$\hat{b}_{1,k}(s_1^0) = v_{1,k}(s_1^0, s_2^0) = v_{2,m-k+1}(s_1^0, s_2^0) = \hat{b}_{2,m-k+1}(s_2^0). \tag{6.1}$$

Our assumption (iv) allows us to define the bidding functions. To see why, consider a unit k, signal s. By (iv), there is a unique signal for bidding 2 such that

$$v_{1,k}(s_1, \alpha) = v_{2,m-k+1}(s_1, \alpha).$$

By equation (6.1), we should define for every good k and signal s

The α corresponds to s_2^0 when $s = s_1^0$.

$$\hat{b}_{1,k}(s_1) = v_{1,k}(s_1, \alpha).$$

Similarly, we can find for any k and s_1 the unique β such that

$$v_{2,k}(\beta, s_2) = v_{1,m-k+1}(\beta, s_2)$$

and set

$$\hat{b}_{2,k}(s_2) = v_{2,k}(\beta, s_2).$$

We now have candidate equilibrium bidding functions. Note that when $\hat{b}_{1,k}(s_1) = v_{1,k}(s_1, \alpha)$, then α depends on k and s_1. By our assumptions, we see that $\alpha(s_1, k)$ is increasing in s_1 and decreasing in k: as s_1 increases, assumption (iii) implies $v_{1,k}$ grows faster than $v_{2,m-k+1}$, pushing the crossing α upward; as k increases, $v_{1,k}$ falls (by (i)) while $v_{2,m-k+1}$ rises, pushing α downward. Similarly, $\beta(s_2, k)$ is increasing in s_2 and decreasing in k. Hence, $\hat{b}_{i,k}(s_i)$ is strictly increasing in s_i. Nice! Furthermore,

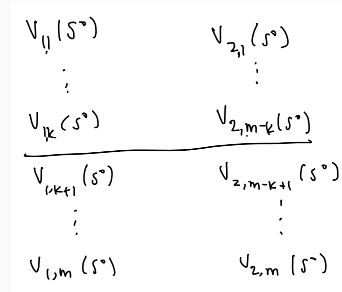
$$\hat{b}_{i,1}(s_i) \geq \cdots \geq \hat{b}_{i,m}(s_i).$$

Proposition 6.5.2 (Efficiency of the equilibrium bidding functions).

The candidate bidding functions $\hat{b}_{1,k}$ and $\hat{b}_{2,k}$ defined above yield an efficient allocation for every signal realization.

Proof of Proposition 6.5.2. We must show that these functions yield an equilibrium and an efficient outcome. We show the latter first. Let s_1^0, s_2^0 be a pair of signals such that it would be efficient for bidder 1 to receive k units and bidder 2 to receive $m-k$ units. Notate $s^0 := (s_1^0, s_2^0)$. Thinking or by referring to the image in the margin, we know

Figure 6.3. *Partition of signals.*



$$v_{1,k}(s_1^0, s_2^0) > v_{2,m-k+1}(s_1^0, s_2^0) \tag{6.2}$$
$$v_{2,m-k}(s_1^0, s_2^0) > v_{1,k+1}(s_1^0, s_2^0). \tag{6.3}$$

Equation (6.2) says bidder 1's marginal value for the kth unit exceeds bidder 2's value for the $(m-k+1)$th, so the kth unit belongs to bidder 1 in an efficient allocation; equation (6.3) is the symmetric statement for bidder 2's $(m-k)$th unit.

We must show that if the two bidders use their $\hat{b}$ bidding functions, then when their signals are s_1^0 and s_2^0, bidder 1 wins exactly k units. Since

$$\hat{b}_{1,k}(s_1^0) = v_{1,k}(s_1^0, \alpha) = v_{2,m-k+1}(s_1^0, \alpha),$$

equation (6.2) and assumption (iii) imply that $\alpha > s_2^0$. Thus

The gap

$v_{1,k}(s_1^0, s_2^0) - v_{2,m-k+1}(s_1^0, s_2^0) > 0$

by (6.2), and by assumption (iii) it is decreasing in s_2. So the zero of the gap, α, must lie above s_2^0.

$$\hat{b}_{1,k}(s_1^0) = v_{1,k}(s_1^0, \alpha) \geq v_{1,k}(s_1^0, s_2^0),$$

where the inequality uses $\alpha \geq s_2^0$ and that $v_{1,k}$ is nondecreasing in s_2 (assumption (ii)). Similarly, since

$$\hat{b}_{2,m-k+1}(s_2^0) = v_{2,m-k+1}(\beta, s_2^0) = v_{1,k}(\beta, s_2^0),$$

equation (6.2) and assumption (iii) give for $\beta < s_1^0$

$$\hat{b}_{2,m-k+1}(s_2^0) = v_{2,m-k+1}(\beta, s_2^0) \leq v_{2,m-k+1}(s_1^0, s_2^0). \tag{6.4}$$

Chaining these two bounds with equation (6.2) gives

$$\hat{b}_{1,k}(s_1^0) \geq v_{1,k}(s^0) > v_{2,m-k+1}(s^0) \geq \hat{b}_{2,m-k+1}(s_2^0),$$

so $\hat{b}_{1,k}(s_1^0) > \hat{b}_{2,m-k+1}(s_2^0)$: bidder 1 wins the kth unit. We symmetrically get from equation (6.3) and assumption (iii) that

$$\hat{b}_{2,m-k}(s_2^0) \geq v_{2,m-k}(s_1^0, s_2^0) > v_{1,k+1}(s_1^0, s_2^0) \geq \hat{b}_{1,k+1}(s_1^0), \tag{6.5}$$

so $\hat{b}_{2,m-k}(s_2^0) > \hat{b}_{1,k+1}(s_1^0)$: bidder 1 does not win the $(k+1)$th unit.

Equations (6.4) and (6.5) are exactly what determine an efficient outcome, so the outcome is efficient regardless of the signals. □

Example 6.5.3.

Consider $m = 2$, where

$$v_{1,1}(s_1, s_2) = 30s_1 + 20s_2, \quad v_{1,2}(s_1, s_2) = 20s_1 + 10s_2$$

and

$$v_{2,1}(s_1, s_2) = 12s_1 + 50s_2, \quad v_{2,2}(s_1, s_2) = 10s_1 + 40s_2.$$

To find $\hat{b}_1(\cdot)$ and $\hat{b}_2(\cdot)$, we want to find α [a]such that

$$\hat{b}_{1,1}(s_1) = v_{1,1}(s_1, \alpha) = v_{2,2}(s_1, \alpha),$$

i.e. solving

$$30s_1 + 20\alpha = 10s_1 + 40\alpha,$$

which yields $\alpha = s_1$, so $\hat{b}_{1,1}(s_1) = 50s_1$. We do the same for bidder 1's second unit (and bidder 2's first): find α such that

$$\hat{b}_{1,2}(s_1) = v_{1,2}(s_1, \alpha) = v_{2,1}(s_1, \alpha),$$

i.e. solving

$$20s_1 + 10\alpha = 12s_1 + 50\alpha,$$

which yields $\alpha = \frac{1}{5}s_1$, so $\hat{b}_{1,2}(s_1) = 22s_1$.
Symmetrically, we find β such that

$$\hat{b}_{2,1}(s_2) = v_{2,1}(\beta, s_2) = v_{1,2}(\beta, s_2),$$

i.e.

$$12\beta + 50s_2 = 20\beta + 10s_2,$$

so $\beta = 5s_2$ and $\hat{b}_{2,1}(s_2) = 110s_2$. Lastly, we find β such that

$$\hat{b}_{2,2}(s_2) = v_{2,2}(\beta, s_1) = v_{1,1}(\beta, s_2),$$

i.e.

$$1 - \beta + 40s_2 = 30\beta + 20s_2.$$

We find $\beta = s_2$ and $\hat{b}_{2,2}(s_2) = 50s_2$.
If $s_1 = 1$ and $s_2 = 2$, then

$$\hat{b}_{1,1}(1) = 50,\ \hat{b}_{1,2}(1) = 22,\ \hat{b}_{2,1}(2) = 220,\ \hat{b}_{2,2}(2) = 100.$$

So bidder 2 bids 22 for the first unit and 50 for the second unit, totalling 72. Bidder 2 would not like to give up her second unit nor her second unit and her first, and bidder 1 would not like to win a first unit, nor a first unit and a second unit, so the outcome is an equilibrium. It is also efficient.

[a] We know such an α always exists by our assumptions, which gave us this procedure of finding the bidding functions.

Collusion 7

We briefly discuss collusion in the context of strong cartels, which allow for transfer payments between members of the bidding ring.

7.1 Strong Cartels

We consider an independent private values model with symmetric values $v_i \sim f(v_i)$ and $v_i \in [0,1]$ and all N members participating in the bidding ring. Suppose the seller uses a first-price auction with reserve price ρ.

We make no restriction on how ρ is determined.

The best the bidding ring can do win the good when $\max\{v_1, \dots, v_N\} \geq \rho$, in which case they pay the seller ρ. The bidders do this by running a "knockout" auction among themselves.

Definition 7.1.1 (Knockout auction).

In a knockout auction, the N bidders in the bidding ring submit a sealed bid $b_i \in [0, \infty)$. The highest bid wins and the corresponding bidder earns the right to bid ρ in the real auction, where all others must bid less than ρ in the real auction. The high bidder keeps the good, and pays each of the other bidders the side payment $\frac{b-\rho}{N-1}$.

The total payment made by a winner of the knockout auction (and the main auction) is $\rho + (N-1)\frac{b-\rho}{N-1} = b$. We see that this format is equivalent to a first price auction, except that losers get a side payment.

Suppose $\tilde{b}(\cdot)$ is a symmetric equilibrium bidding function for the knockout auction. Assume that $\tilde{b}$ is strictly increasing, with $\tilde{b}(v) = 0$ for $v < \rho$ and $\tilde{b}(\rho) = \rho$. We want to determine what $\tilde{b}$ is above ρ. Let $u(r,v)$ be a bidder's utility when her value is v and she bids $\tilde{b}(r)$ and all others bid according to $\tilde{b}(\cdot)$. We write down

$$u(r,v) = F^{N-1}(r)\left(v - \tilde{b}(r)\right) \tag{7.1}$$

$$+ \int_r^1 \underbrace{\left(\frac{\tilde{b}(s)-\rho}{N-1}\right)}_{\text{side payment}} \overbrace{(N-1)\,F^{N-2}(s)f(s)}^{\text{density of highest bid among others}}\, ds. \tag{7.2}$$

Maximizing as we have done, for $v \geq \rho$,

$$\frac{du}{dr} = (N-1)F^{N-2}(r)f(r)\left(v - \tilde{b}(r)\right) - F^{N-1}(r)\tilde{b}'(r) - \left(\tilde{b}(r) - \rho\right)F^{N-1}(r)f(r).$$

The equilibrium condition $\left.\frac{du}{dr}\right|_{r=v} = 0$ yields a linear ODE for $\tilde{b}$. Multiplying by $F(v)$ and collecting terms gives

$$G(v) = F^N(v)\left(\tilde{b}(v) - \rho\right).$$

Then

$$G'(v) = (N-1)F^{N-1}(v)f(v)(v-\rho).$$

Integrating from ρ to v yields

$$F^N(v)\left(\tilde{b}(v) - \rho\right) = \int_\rho^v (N-1)F^{N-1}(x)f(x)(x-\rho)dx.$$

Since $\tilde{b}(\rho) = \rho$, this simplifies to

$$\tilde{b}(v) - \rho = \frac{N-1}{N}m(v).$$

where

$$m(v) = \mathbb{E}\left[X_{(1)} - \rho \mid X_{(1)} < v\right].$$

We can verify this $\tilde{b}$ is indeed increasing in v and constitutes an equilibrium (by examining the sign of $\frac{du}{dr}$).